Pest Detection Information Front-End

Team project plan | SDG 2: Zero Hunger | Proposed scope and responsibilities | 9 October 2026

Project summary. A responsive, accessible information website that helps farmers and growers explore common crop pests through photographs, descriptions and an interactive symptom-based identification guide. The guide suggests possible matches; it is not an AI image-recognition or definitive diagnostic tool.

1 | Problem, users and sustainability purpose

Farmers and growers need clear, usable information when they notice damage or insects on crops. A well-organized identification resource can help users narrow down possible pests, understand characteristic symptoms and find trustworthy next steps. The project aligns with UN SDG 2 (Zero Hunger) through crop protection, agricultural knowledge and potentially reduced avoidable losses; it does not claim to measure real-world yield improvements.

Target users: small-scale farmers, growers, allotment users and students. Primary use case: select a crop and visible symptoms, receive a shortlist of possible pests, then read evidence-based information and follow links to authoritative guidance.

2 | Minimum viable product (MVP)

Feature	What the user can do
Pest gallery	Browse 8-12 carefully researched pests with clear, attributed images.
Search and filters	Filter by crop, pest name or symptom; reset filters.
Pest detail view	Read identifying features, affected crops, typical damage, life-cycle context and safe next steps.
Interactive identification guide	Select crop and observed symptoms; JavaScript returns possible matches with reasons, not a certain diagnosis.
Responsive and accessible UI	Use the website on mobile and desktop with keyboard-friendly navigation and meaningful image alt text.

Stretch goals only if the MVP is complete: compare two pests, a glossary, saved favourites using localStorage, or printable pest summaries. No backend, login, live image recognition, or pesticide recommendation engine is required.

3 | Proposed technology and architecture

HTML for semantic structure; CSS for responsive layouts; JavaScript for filtering, search, detail views and the decision guide; a local JSON file or JavaScript data object for curated pest records; GitHub for version control and submission. This directly reflects the guide's frontend emphasis. MySQL and Python are not necessary for this version.

Data flow: curated pest dataset -> JavaScript filtering/matching -> results cards -> pest detail view. Each pest record should include ID, common/scientific name, crops, symptoms, identifying features, image credit, guidance and source URLs.

Research, differentiation and delivery

4 | Research and credibility

The group should compare at least 2-3 existing agricultural pest-information services and identify strengths, weaknesses and usability gaps. Suggested research starting points: RHS (rhs.org.uk), AHDB (ahdb.org.uk), UK Plant Health Information Portal (planthealthportal.defra.gov.uk), and FAO (fao.org). These are leads to investigate, not a claim that their content has been fully verified for this proposal.

For every pest, record the original source, publication/access date, crop relevance, symptoms and permitted image usage. Prefer openly licensed or expressly permitted images with attribution. Avoid copying text or photographs without permission. For potentially harmful control methods, link to current authoritative guidance rather than inventing chemical treatments.

5 | What makes this version distinctive?

Proposed differentiator: a simple, accessible crop + symptom guided search designed for quick use in the field. The system explains why each possible match appears and encourages users to compare images and descriptions. This is a usability-led design proposition, not a claim that no similar tool exists.

Example: user selects tomato and holes in leaves; the system shows relevant possible pests, a matching-symptom explanation, and a reminder to verify identification. The team must research the actual pest-symptom mappings before populating this example in the working app.

6 | Proposed five-person team structure

Member / role	Deliverables
1 - Data and identification logic lead (you)	Define pest JSON schema, curate and validate entries with research lead, implement/test JavaScript matching rules.
2 - Frontend layout lead	HTML page structure, navigation, responsive CSS and shared components.
3 - UI/UX and gallery lead	Wireframes, image gallery, detail page, accessibility and usability improvements.
4 - Research and content lead	Competitor analysis, credible pest descriptions, citations, image permissions and SDG 2 justification.
5 - Integration and QA lead / Scrum Master	GitHub coordination, integration, test plan, issue tracking, documentation and sprint facilitation.

Shared responsibilities: all members contribute to backlog decisions, stand-ups, peer review, testing, presentation and documentation. The lecturer may act as Product Owner according to Week 1 teaching; confirm this with the tutor. Names and roles require group approval.

7 | Agile Scrum and project evidence

Use a prioritized product backlog, short weekly sprints, a visible task board (To do / In progress / Review / Done), brief stand-ups, sprint reviews and retrospectives. Record attendance, decisions, blockers and completed work. Keep a Gantt chart, risk log, requirements list, use-case diagram, DFD, wireframes and testing evidence as appropriate to the assessment.

Schedule and assessment plan

8 | Four-week build plan and milestones

Period	Team tasks	Milestone
Week 1 | 5-11 Oct	Confirm concept and roles; complete F1; compare competitors; agree crops/pests, user requirements, GitHub and backlog.	Approved scope and research plan.
Week 2 | 12-18 Oct	Research 8-12 pests; confirm sources/image rights; wireframes, data schema, UI skeleton and guide logic design.	Clickable skeleton and approved content structure.
Week 3 | 19-25 Oct	Implement gallery, search/filter, detail views and guided matching; integrate content and run early tests.	Working end-to-end identification journey.
Week 4 | 26 Oct-1 Nov	Accessibility, responsive checks, user testing, bug fixes, README, diagrams and GitHub submission preparation.	Stable tested prototype by 1 Nov.
Week 5 | 2-8 Nov	Construction/source code due 2 Nov; rehearse demo/viva; refine presentation and individual academic article.	Viva-ready; final materials for 9 Nov.

9 | Assessment deliverables

Assessment	Weight	What to prepare
S1 - Group construction	30%	Working frontend artefact and source code on GitHub by 2 Nov.
S1 - Live viva / slides	20%	Explain research, Scrum/SDLC, use cases, DFD, UI, implementation, testing and demo.
S1 - Individual article	25%	Approx. 1,500 words (+/-10%); original analysis and IEEE-style references, due 9 Nov.
S2 - Individual portfolio	25%	Approx. 200 words (+/-10%) per required entry; reflect on real tasks, feedback, skills and evidence.

Submission check: The assessment guide indicates S2 entries for Weeks 2-5, while teaching slides mention Week 1. Confirm exact Moodle dates and requirements. The Week 1 F1 formative exercise is separate: a collaboratively prepared 5-10 slide presentation OR Word document of no more than 2 pages and 250 words; each student uploads their own copy.

10 | Risks and mitigation

Risk	Mitigation
Pest facts or images inaccurate / unlicensed	Research lead validates sources and permissions; attribution recorded in dataset.
Guide implies a definitive diagnosis	Label results as possible matches; show uncertainty and verification advice.
Scope creep into AI or backend	Freeze MVP in Week 2; optional features only after all essentials pass testing.
Integration or merge conflicts	Use small GitHub branches/pull requests and integrate weekly.
Poor mobile/accessibility experience	Test keyboard, alt text, contrast and mobile layouts throughout development.

Next team meeting: decisions and actions

11 | Decisions to make now

• Approve the project title, audience and SDG 2 rationale.

• Choose a limited set of crops and 8-12 pests; agree inclusion criteria.

• Confirm the interactive crop-and-symptom guide as an MVP feature.

• Agree to a frontend-only implementation with HTML, CSS, JavaScript and JSON.

• Assign names to the five roles and confirm Scrum Master.

• Agree research sources, image attribution process and how to review factual accuracy.

• Create GitHub repository, task board, shared evidence folder and first sprint backlog.

12 | First sprint tasks by responsibility

Owner	Immediate task
Data / logic	Draft a pest-record schema and 3 sample records; sketch crop/symptom matching rules.
Frontend	Create basic HTML pages, navigation and CSS file structure.
UI/UX	Sketch mobile and desktop layouts for gallery, detail and identification guide.
Research	Find 2-3 competitors; shortlist credible pest sources and image licences.
Integration / Scrum	Set up repository, README, Kanban board, meeting notes and test checklist.

13 | Individual S2 reflection prompts

At the end of each day, note: what I actually did; what I learned; what was difficult and how I responded; how my contribution affected the team; and what I will do next. Save screenshots, Git commits, research notes, wireframes, meeting decisions and feedback. Write the weekly reflection from these genuine notes in your own voice.

14 | Demo scenario for the viva

A user opens the site on a mobile phone, chooses a crop, selects visible symptoms, sees a shortlist of possible pests, opens one pest profile, compares the images and distinguishing features, and follows a trusted source for further verification. Demonstrate filtering, responsive layout and clear uncertainty wording.

15 | Source and scope notes

University documents: CPU4106-20C Group Project Topic Guide, page 3 (Project 20, HTML/CSS/JavaScript/Image Gallery UI, SDG 2); Formative Assessment 1 - Group Project Reflection and Planning (five planning questions and individual Moodle upload). The assessment weights and deadlines are based on the previously shared module assessment materials. This document is a proposed plan, not a confirmation of final team roles or tutor approval.

Research leads: UN SDG 2: https://sdgs.un.org/goals/goal2 ; RHS: https://www.rhs.org.uk ; AHDB: https://ahdb.org.uk ; UK Plant Health Portal: https://planthealthportal.defra.gov.uk ; FAO: https://www.fao.org . Verify the suitability of individual pages before citing them in submitted work.

Team pitch: “We will build a responsive, accessible pest-information website with an interactive crop-and-symptom guide. The site will help growers narrow down possible pest matches, compare reliable images and read researched guidance, demonstrating HTML, CSS, JavaScript, UI/UX, research and Agile teamwork.”